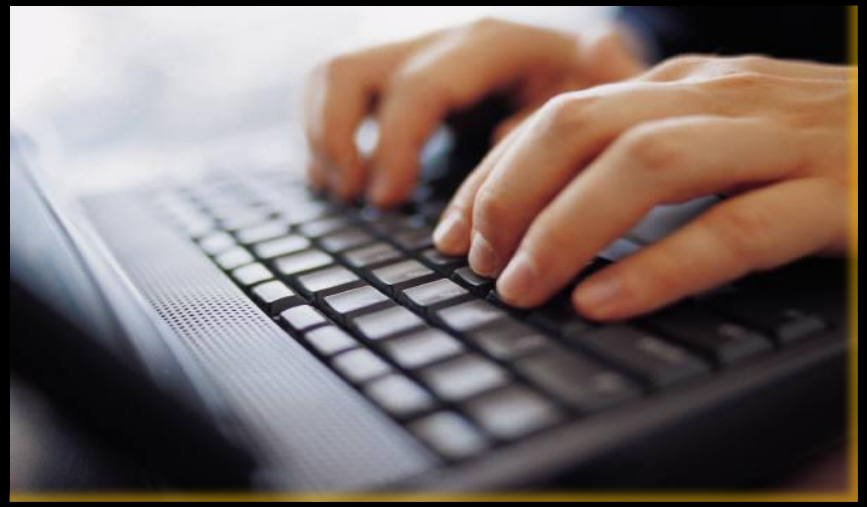
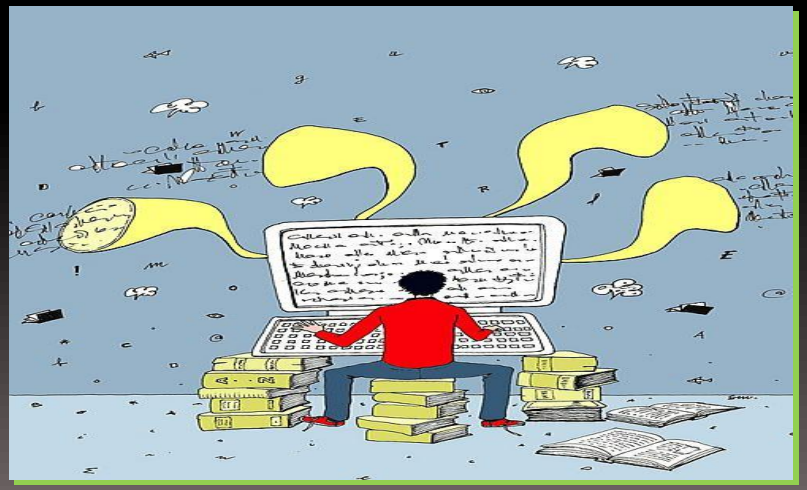


MOBILE COMPUTING

An Seminar By ... GAURAV KORIYA

I T II YEAR

Table Of Contents



- Introduction
- What is Mobility
- Mobile Computing Device
- Wireless Network
- Security and Privacy
- Disconnected Operations
- Future Aspects

Introduction

- Mobile Computing

  Mobile (Moving)

  +

  Computing (using computer)

- **Mobile Computing**

is an umbrella term used to describe technologies that enable people to access network services anyplace, anytime, and anywhere

Mobile computing

- In recent years, distributed computer has become the norm, especially after the advent of the internet.
- Today, technological advances are shaping a new computing environment where the user is free from the need to work at a specific location, and is even enabled to work while on the move.
- This new paradigm is often called MOBILE COMPUTING.

What Is Mobility ?

- **A device that moves**
  - Between different geographical locations
  - Between different networks
  - Between different applications
  - Such as Palm-top, laptop, Cell phones
- **A device due to which mobile communication is possible**
  - Modems.

MOBILE COMPUTING DEVICE

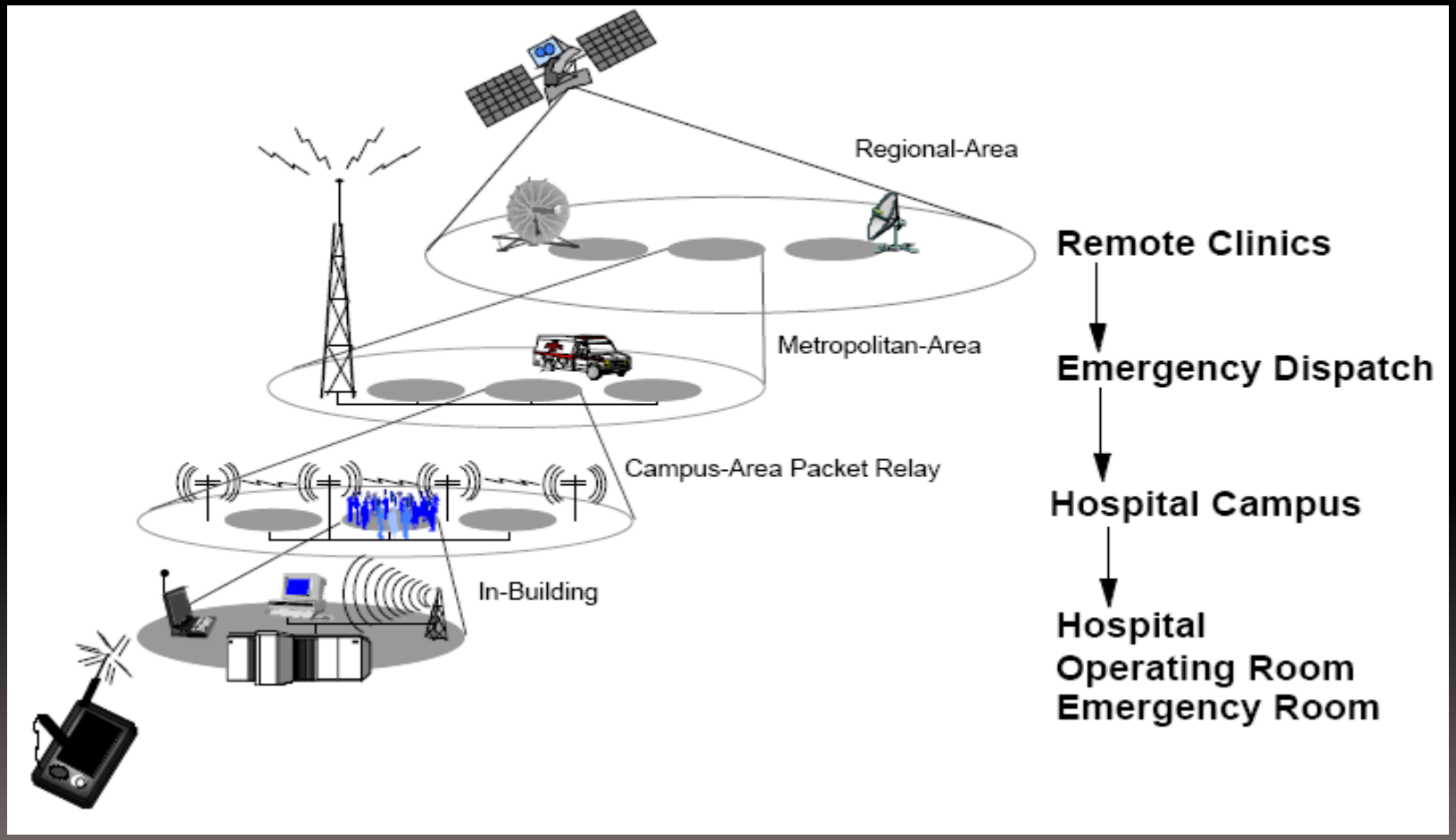
Mobile Computing Devices

- **Display Only**
  - Info Pad model: limited portable processing
  - Like a cordless phone.
- **Laptop Computer**
  - ThinkPad model: significant portable processing, operates independently of wireless infrastructure.
- Personal Digital Assistant **[PDA]** :Somewhere between these extremes. a handheld device that combines computing, telephone/fax, Internet and networkingfeatures.

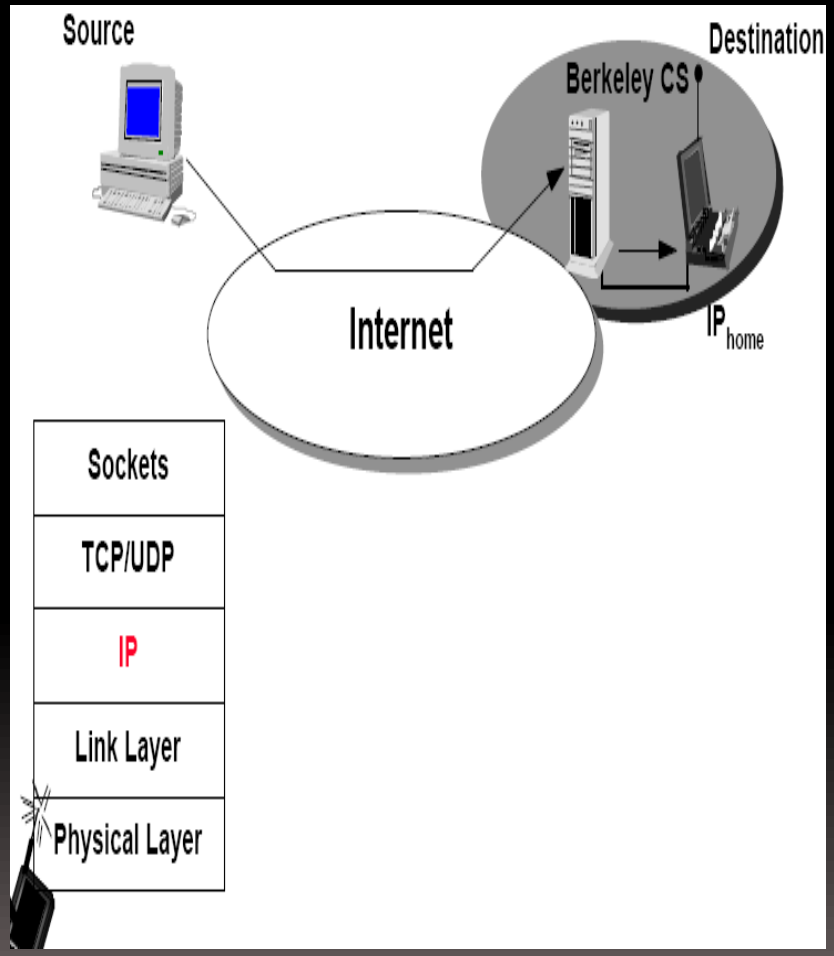
WIRELESS NETWORK TECHNOLOGY

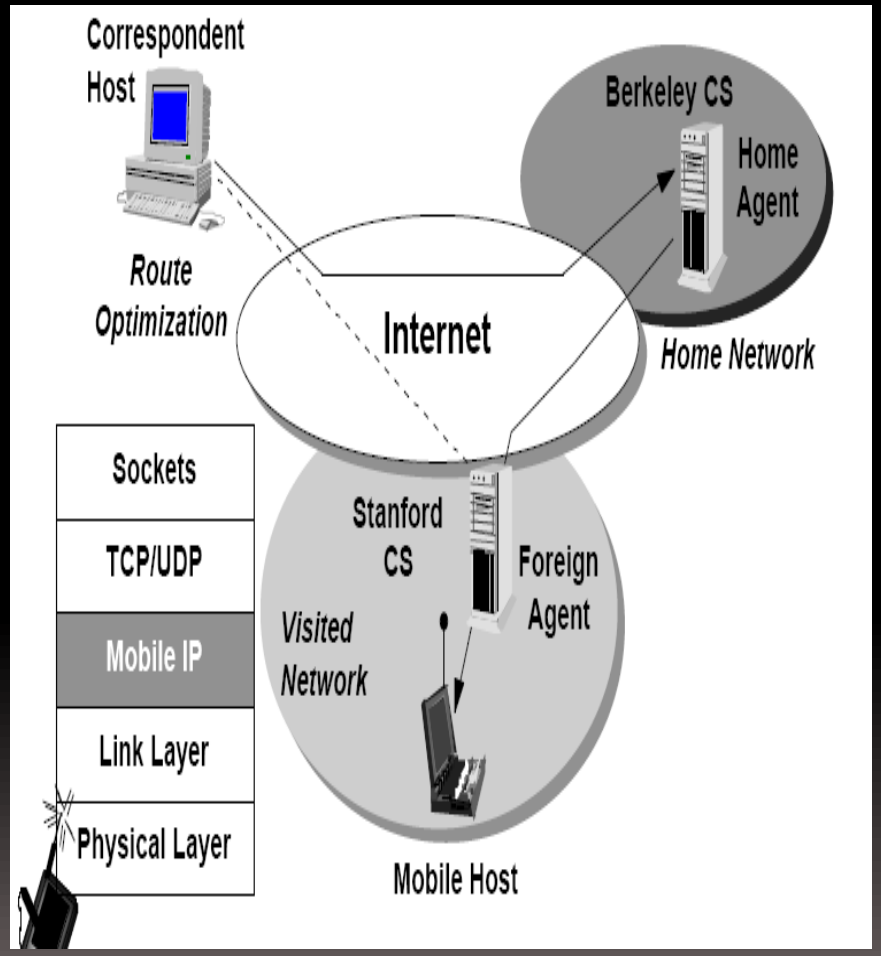
Wireless Network Technology : Inside It

"Wireless Overlay" Network

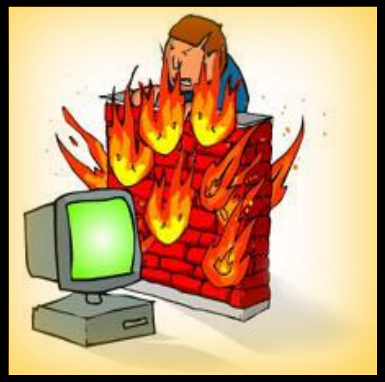


Mobile IP


Source
Destination
Berkeley CS
Internet
IP home
Sockets
TCP/UDP
IP
Link Layer
Physical Layer


Correspondent Host
Berkeley CS
Home Agent
Route Optimization
Internet
Home Network
Sockets
TCP/UDP
Mobile IP
Link Layer
Physical Layer
Stanford CS
Foreign Agent
Visited Network
Mobile Host

Wireless Networks



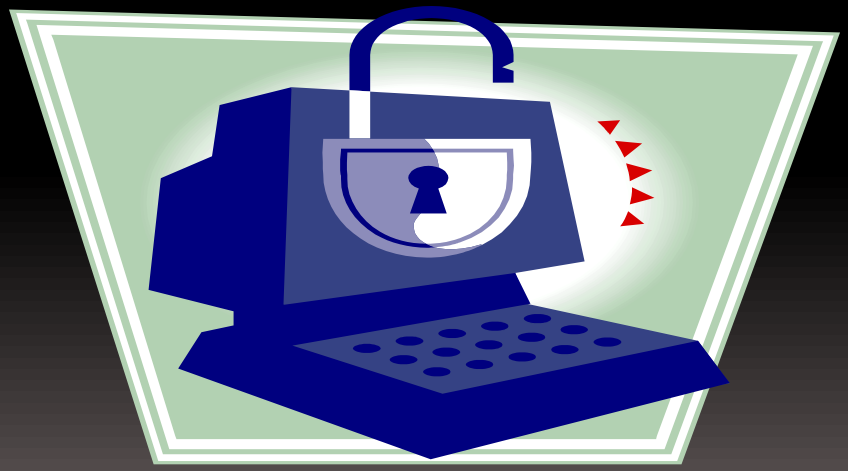
SECURITY AND PRIVACY

Security and Privacy

- **Security and privacy Threats:**

Wi-Fi networks are a shared network that makes it easier for others to eavesdrop on your communication.

- Devices are Stolen and tampered.
- Cryptography.
- Connection.



❑ Security and privacy Solutions:

- Secure Web Browsing
  - Use secure, encrypted sessions.
- Secure Internet Transactions
  - Use UCI's VPN to encrypt your network traffic.
- Always use a Personal Firewall
  - Set the firewall to deny ALL incoming connections.

- Detecting man in the middle trying to steal information
  - Each device displays the hash of its public key.
  - The keys are exchanged by Diffie-Hellman protocol.
  - The two hashes of the keys are compared.
  - If the two match then there is no man in the middle.
  - If the two don't match it's assumed there is a man in the middle.

Never store Sensitive Data on mobile devices unless absolutely necessary.

Disconnected Operations
AND
Various Challenges

Disconnected Operations

- Low bandwidth
- High bandwidth variability
- Low power and resources
- Security risks
- Wide variety terminals and devices with different capabilities
- Fit more functionality into single, smaller device
- Disconnection

LIMITATIONS

Insufficient bandwidth

If the user needs access to a network they must resort to slow wireless WAN systems primarily intended for telephone use. Higher speed wireless LANs are only available in specific sites

Security standards

When working mobile one is dependent on public networks, requiring careful use of VPNs.

Power consumption

Mobile computers must rely entirely on battery power. Combined with compact size, this means unusually expensive batteries be used

Transmission interferences

Weather and terrain problems as well as distance-limited connection exist with some technologies. Reception in tunnels and some buildings is poor.

Potential health hazards

Potential health damage from cellular radio frequency emission is not known yet. However, more car accidents are related to drivers who were talking through a mobile device. Also, cell phones may interfere with sensitive medical devices.

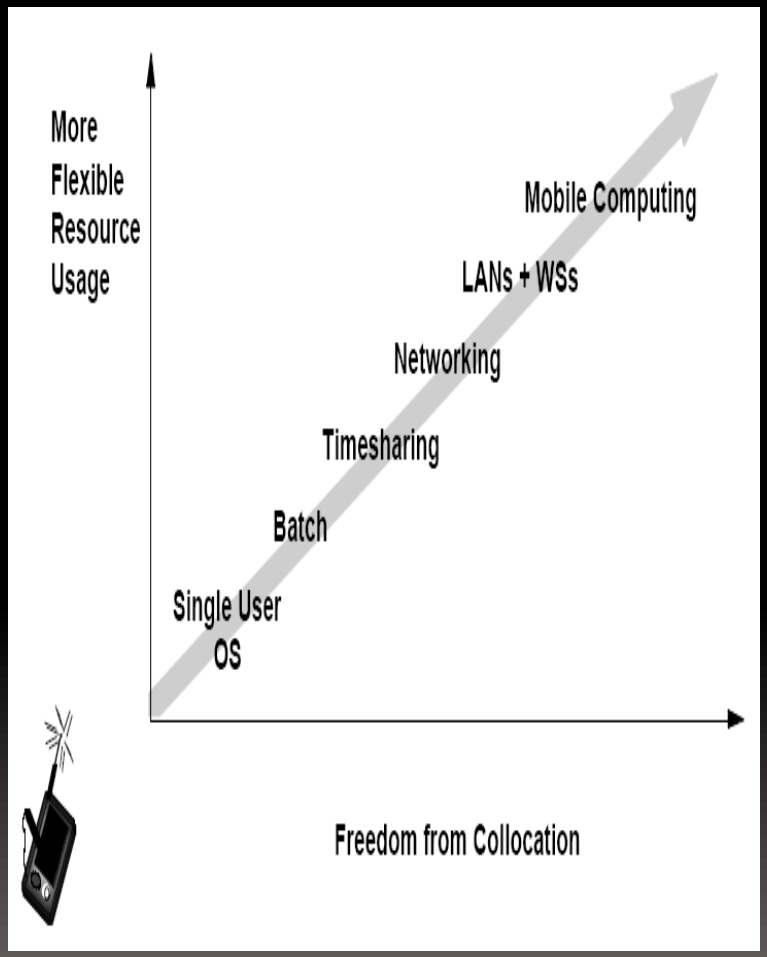
Human interface with device

As, screens are often too small. Keyboards are impractical, especially one-handed, and alternate methods such as speech or handwriting recognition require training.

Future Aspects

- Track your friends on a map by picking up GPS signals from other Android phones, so you can search for nearby parties.
- Geolife

This app will send you reminders based on location instead of time. For example, as you approach a grocery store, it will remind you to pick up milk.



Conclusion

- Mobile computing has severe limitations

- however, it is far from impossible, and technology improves all the time

- Lots of challenges

- some have (good) solutions, many others are still waiting to be solved

Thank You……